

COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

Midsemester Exam #2

Only one sheet of paper 5 x 7 is allowed

Duration: 55 minutes --- March 9, 2020

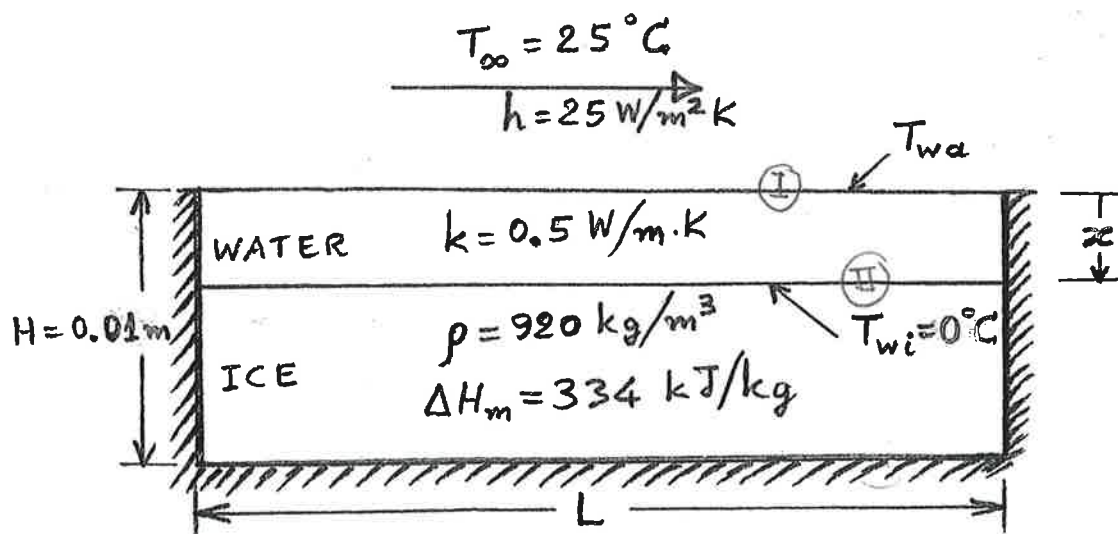
60/125

Problem 1: (50 points)

A slab of ice in a container of dimensions L , L , and H is placed on a well-insulated pad. At its top surface, initially the ice is exposed to ambient air for which $T_\infty = 25^\circ\text{C}$ and the convection coefficient is $25\text{ W/m}^2\cdot\text{K}$. The ice is melting and the water formed acts as an additional resistance to the heat transfer. Assume that there is no storage of energy in the water being formed and that the water temperature profile is linear. The ice temperature remains at 0°C and $H = 10\text{ mm}$. If the density and latent heat of fusion of ice are 920 kg/m^3 and 334 kJ/kg , respectively, and the thermal conductivity of water is $0.5\text{ W/m}\cdot\text{K}$, then:

- By performing energy balance(s) over appropriate control element(s) determine the position of the water/ice interface as a function of time.
- Determine the time required for the ice to melt. How does this relate to the time found in the case of no water-layer resistance?

Bonus: (25 points) Determine how the water-air interface temperature, T_{wa} , changes with time. What is the value of T_{wa} when the ice has melted?



①

Energy balance

$$a) \quad In - out + gen - consumption = 0$$

$$In = out$$

Energy balance at position I

$$\dot{Q} = \frac{dQ}{dt} (m \Delta H_m)$$

$$\frac{q}{A} = \bar{h}_l (T_\infty - T_s) \Big|_{In} = k (T_2 - T_1) \Big|_{out} \times$$

$$25(x) [25 - T_{wa}] \Big|_{In} = 0.5 (T_{wa} - T_{wi}) \Big|_{out}$$

$$25x [25 - T_{wa}] \Big|_{In} = 0.5 (T_{wa} - T_{wi}) \Big|_{out}$$

$$625x - 25T_{wa} \Big|_{In} = 0.5 T_{wa} \Big|_{out} \rightarrow \textcircled{1}$$

Energy balance at position II

$$\frac{q}{A} = k (T_2 - T_1) \Big|_{In} = \bar{h} \Delta T_{lm} \Big|_{out}$$

$$\frac{q}{A} = k (T_2 - T_1) \Big|_{In} = \bar{h}_l (T_s - T_\infty) \Big|_{out}$$

$$= 0.5 (T_{wi} - T_1) \Big|_{In} = \frac{324(0.01-x)}{920} (T_{wi} - T_\infty) \Big|_{out} \quad \begin{matrix} T_{wi} = 0^\circ C \\ T_\infty = 25^\circ C \end{matrix}$$

$$= 0.5 (-T_1) \Big|_{In} = 0.3630 (0.01-x) (-25^\circ C) \Big|_{out}$$

$$= -0.5 T_1 \Big|_{In} = \{0.003630 - 0.3630x\} - 25$$

$$= -0.5 T_1 \Big|_{In} = -0.0907 + 0.9075x \Big|_{out} \rightarrow \textcircled{2}$$

x+0.01 combining equation ① and ②

$$t = \int_0^{x+0.01} 625x - 25T_{wa} - 0.5T_1 + 0.5T_{wa} - 0.0907 + 0.9075x$$

b) time required for the ice to melt would be determined by performing the energy balance from ice $\rightarrow$ water.

$$\frac{q}{A} = \bar{h} L (T_s - T_{\infty}) \Big|_{In} = k (T_2 - T_1) \Big|_{out}$$

$$= \frac{334}{920} (0.01 - x) (T_{wi} - T_{\infty}) \Big|_{In} = 0.5 (T_{wi} - T_1) \Big|_{out}$$

$$= 0.3630 (0.01 - x) (-25^{\circ}\text{C}) \Big|_{In} = 0.5 (-T_1) \Big|_{out}$$

$$= -0.0907 + 0.9075x \Big|_{In} = -0.5 T_1 \Big|_{out}$$

$$t = \int_0^{x+0.01} -0.0907 + 0.9075x + 0.5 T_1 \quad \times$$

Problem 2: (75 points)

A fluid at 100 °C is to be cooled by a different stream of the same fluid at 20 °C. A heat exchanger is available for parallel or counter flow. When the heat exchanger is used with parallel flow, the outlet temperature of the hot stream is 65 °C. If the mass flow rates in the hot and cold streams are equal to each other and the heat capacity of the fluid is temperature independent, determine:

- (a) (30 points) The outlet temperature of the cold stream with parallel flow in the heat exchanger, and
(b) (45 points) The outlet temperatures of the cold and hot streams when a counter flow set up is used.

g) $q = UA(T_a - T)$

Parallel: $NTU = \frac{-\ln[1 - \epsilon(1 + C_R)]}{1 + C_R}$

$m_{h,h} = m_{h,c}$
carnet out

$$C_R = \frac{C_{min}}{C_{max}} = \frac{1}{1 + C_R}$$

$$\dot{m}_c (T_2 - T_1) = \dot{m}_h (T_1 - T_2)$$

$$\dot{m}_c = \dot{m}_h$$

$$c_h (T_2 - T_1) = c_c (T_1 - T_2)$$
$$c_h (100 - 65) = c_c (T_2 - 20)$$

$$35 = T_2 - 20$$

$$T_{2c} = 55^\circ\text{C}$$

$$NTU = \frac{-\ln[1 - (1 + 0.78)]}{1 + 0.78} = 0.1396$$

b) counter flow

$$NTU = \frac{1}{C_h - 1} \ln \left[\frac{\epsilon - 1}{\epsilon C_h - 1} \right]$$

$$\dot{m}_h (T_2 - T_1) = \dot{m}_c (T_1 - T_2)$$

$$\dot{m}_h = \dot{m}_c$$

$$c_h (T_2 - T_1) = c_c (T_1 - T_2)$$

$$c_h (T_2 - 100) = c_c (20 - T_2) \rightarrow \textcircled{1}$$

$$q = UA (T_2 - T_1)$$

$$q' = U (T_2 - T_1)$$

$$= \frac{1}{1 + C_h} (T_2 - T_1)$$

$$= \frac{1}{1 + 0.1396} (T_2 - T_1)$$

$$(55)(0.1396) = 0.8775 (T_2 - T_1)$$

$$7.678 = 0.8775 (T_2 - T_1) \rightarrow \textcircled{2}$$

$$0.1396 (T_2 - 100) = 0.1396 (20 - T_2) \rightarrow \textcircled{1}$$

$$7.678 = 0.8775 (T_2 - T_1) \rightarrow \textcircled{2}$$

solving for equations 1 and 2 would yield the outlet temperature

T_1 = Hot Stream #

T_2 = Cold Stream #

NOT NECESSARILY NEEDED

TABLE 11.3 Heat exchanger effectiveness relations [5]

Flow Arrangement	Relation	
Parallel flow	$\varepsilon = \frac{1 - \exp[-NTU(1 + C_r)]}{1 + C_r}$	(11.28a)
Counterflow	$\varepsilon = \frac{1 - \exp[-NTU(1 - C_r)]}{1 - C_r \exp[-NTU(1 - C_r)]}$	$(C_r < 1)$
	$\varepsilon = \frac{NTU}{1 + NTU}$	$(C_r = 1)$ (11.29a)
Shell-and-tube		
One shell pass (2, 4, ... tube passes)	$\varepsilon_1 = 2 \left\{ 1 + C_r + (1 + C_r^2)^{1/2} \times \frac{1 + \exp[-(NTU)_1(1 + C_r^2)^{1/2}]}{1 - \exp[-(NTU)_1(1 + C_r^2)^{1/2}]} \right\}^{-1}$	(11.30a)
n shell passes ($2n, 4n, \dots$ tube passes)	$\varepsilon = \left[\left(\frac{1 - \varepsilon_1 C_r}{1 - \varepsilon_1} \right)^n - 1 \right] \left[\left(\frac{1 - \varepsilon_1 C_r}{1 - \varepsilon_1} \right) - C_r \right]^{-1}$	(11.31a)
Cross-flow (single pass)		
Both fluids unmixed	$\varepsilon = 1 - \exp \left[\left(\frac{1}{C_r} \right) (NTU)^{0.22} \{ \exp[-C_r (NTU)^{0.78}] - 1 \} \right]$	(11.32)
$C_{\max}$ (mixed), $C_{\min}$ (unmixed)	$\varepsilon = \left(\frac{1}{C_r} \right) (1 - \exp \{ -C_r [1 - \exp(-NTU)] \})$	(11.33a)
$C_{\min}$ (mixed), $C_{\max}$ (unmixed)	$\varepsilon = 1 - \exp \{ -C_r^{-1} [1 - \exp \{ -C_r (NTU) \}] \}$	(11.34a)
All exchangers ($C_r = 0$)	$\varepsilon = 1 - \exp(-NTU)$	(11.35a)

TABLE 11.4 Heat exchanger NTU relations

Flow Arrangement	Relation	
Parallel flow ✓	$NTU = -\frac{\ln[1 - \varepsilon(1 + C_r)]}{1 + C_r}$	(11.28b)
Counterflow ✓	$NTU = \frac{1}{C_r - 1} \ln \left(\frac{\varepsilon - 1}{\varepsilon C_r - 1} \right)$	$(C_r < 1)$
	$NTU = \frac{\varepsilon}{1 - \varepsilon}$	$(C_r = 1)$ (11.29b)
Shell-and-tube		
One shell pass (2, 4, ... tube passes)	$(NTU)_1 = -(1 + C_r^2)^{-1/2} \ln \left(\frac{E - 1}{E + 1} \right)$	(11.30b)
	$E = \frac{2/\varepsilon_1 - (1 + C_r)}{(1 + C_r^2)^{1/2}}$	(11.30c)
n shell passes ($2n, 4n, \dots$ tube passes)	Use Equations 11.30b and 11.30c with $\varepsilon_1 = \frac{F - 1}{F - C_r}$ $F = \left(\frac{\varepsilon C_r - 1}{\varepsilon - 1} \right)^{1/n}$ $NTU = n(NTU)_1$	(11.31b, c, d)
Cross-flow (single pass)		
$C_{\max}$ (mixed), $C_{\min}$ (unmixed)	$NTU = -\ln \left[1 + \left(\frac{1}{C_r} \right) \ln(1 - \varepsilon C_r) \right]$	(11.33b)
$C_{\min}$ (mixed), $C_{\max}$ (unmixed)	$NTU = -\left(\frac{1}{C_r} \right) \ln [C_r \ln(1 - \varepsilon) + 1]$	(11.34b)
All exchangers ($C_r = 0$)	$NTU = -\ln(1 - \varepsilon)$	(11.35b)

